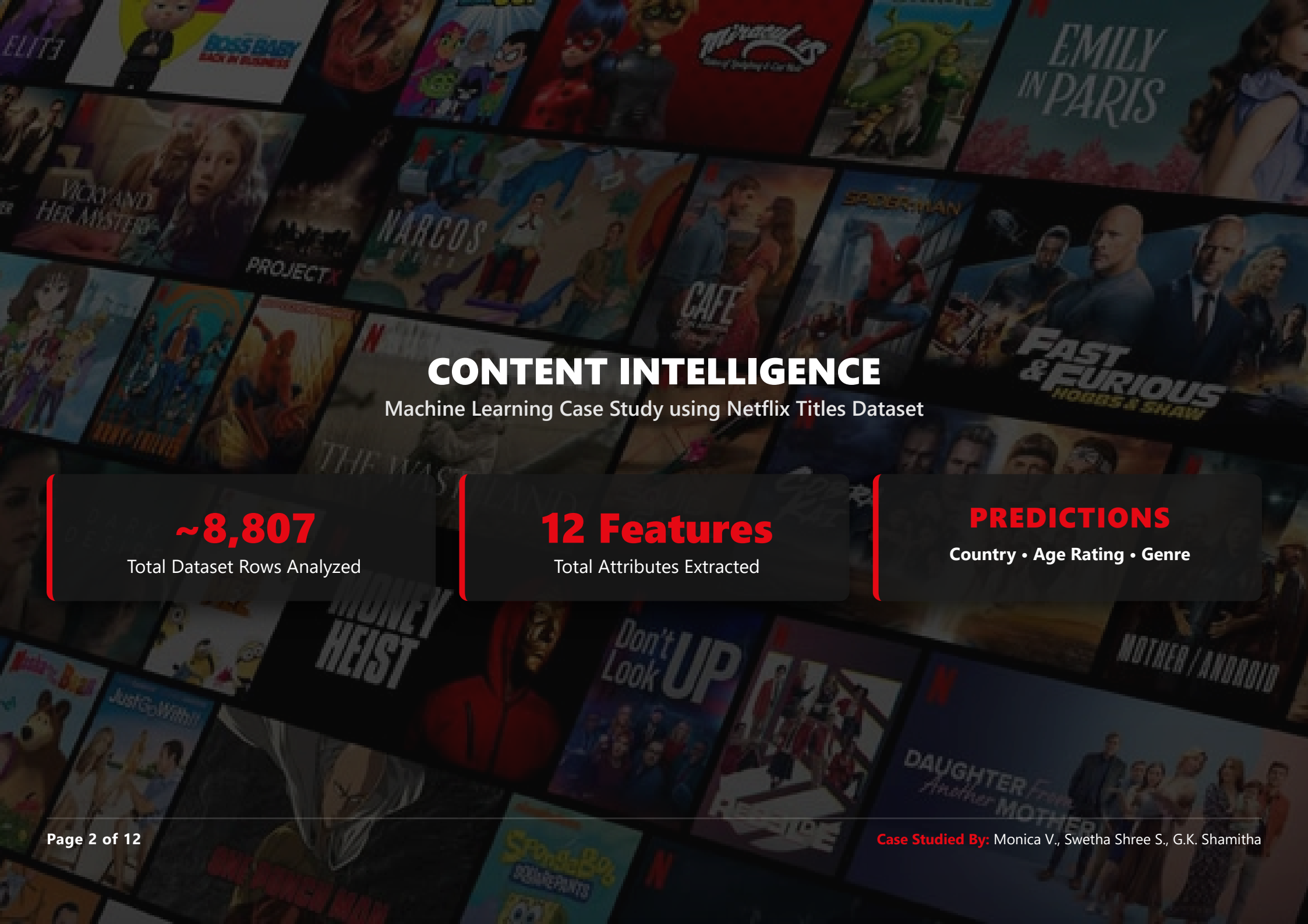
# WELCOME




# CONTENT INTELLIGENCE

Machine Learning Case Study using Netflix Titles Dataset

**~8,807**
Total Dataset Rows Analyzed

**12 Features**
Total Attributes Extracted

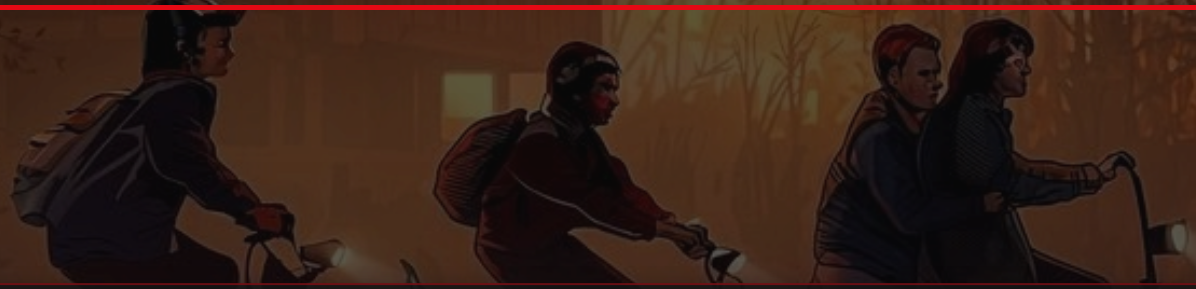
**PREDICTIONS**
**Country • Age Rating • Genre**



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# END-TO-END WORKFLOW ARCHITECTURE (PART 1)

Data Preparation, Recommendation & EDA Stage



**FEATURES:** show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, description

**1** **Data Preprocessing & Feature Engineering**

- Handling Missing Values
- Categorical Encoding
- TF-IDF Vectorization

▼

**2** **Content Recommendation Engine**

- Cosine Similarity Matrix
- Top-5 Content Recommendations Retrieval

▼

**3** **Exploratory Data Analysis (Core Questions)**

- Q1: Country-wise Production Counts
- Q2: Popular Genre Distribution
- Q3: Content Age-Rating Analysis



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# END-TO-END WORKFLOW ARCHITECTURE (PART 2)



**4** **Data Splitting & Model Training Input**

- Train-Test Split Protocol
- Label Encoding & Target Variable Mapping

▼

**5** **Supervised Machine Learning Classifiers**

- Linear SVC
- Logistic Regression
- Random Forest Classifier

▼

**6** **Comprehensive Model Evaluation & Results Export**

- 11 Performance Evaluation Metrics
- Automated Excel Export (df.to_excel)



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## Handling Missing Data

- **Categorical Imputation:** Replaced missing text entries in director, cast, and country with 'Unknown'.
- **Rating Imputation:** Imputed missing rating entries with 'Not Rated'.
- **Null Removal:** Cleared sparse missing values in critical columns like date_added.

## Encoding & Text Processing

- **Label Encoding:** Transformed target categorical fields into machine-readable numerical labels.
- **TF-IDF Vectorization:** Transformed text feature columns into high-dimensional numerical vectors.

## EXAMPLES

### 1. Label Encoding Example

| Feature | Original Categorical | Encoded Label |
|---|---|---|
| country | United States, India, UK | 0, 1, 2 ... |
| rating | TV-MA, TV-14, PG-13 | 0, 1, 2 ... |

### 2. TF-IDF Matrix Representation

| Words \ Movies | Movie 1 | Movie 2 | Movie 3 |
|---|---|---|---|
| love | 0.00 | 0.55 | 0.00 |
| action | 0.65 | 0.00 | 0.50 |
| hero | 0.00 | 0.70 | 0.00 |
| cop | 0.45 | 0.00 | 0.60 |


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## Text Vectorization Setup

- Merged textual columns including description and listed_in (Genres).
- Applied **TF-IDF Vectorizer** to extract essential keywords while ignoring standard stop-words.

## Similarity Matrix Matching

- Calculated **Cosine Similarity** scores across all title description vectors.
- Built a custom query function to retrieve the top 5 most contextually similar Netflix shows/movies.

## REAL-TIME RECOMMENDATION OUTPUT

Input Title Query: **'Squid Game'**

| # | Recommended Title | Type | Genres / Listed In |
|---|---|---|---|
| 1011 | **Free to Play** | Movie | Documentaries |
| 4789 | **King of Peking** | Movie | Comedies, Dramas, International |
| 1615 | **Ink Master** | TV Show | Reality TV |
| 1440 | **Nailed It! Mexico** | TV Show | International TV Shows, Reality TV |
| 2914 | **Isi & Ossi** | Movie | Comedies, International, Romantic |

** Top-5 cosine similarity matches generated during script execution.*



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# EXPLORATORY DATA ANALYSIS (EDA)



## Q1: Country Production

**Which country produces the most Netflix content?**

- Analyzed frequency distribution across production countries.
- United States leads production, followed by India and the United Kingdom.

## Q2: Popular Genres

**Which genres are most popular on Netflix?**

- Aggregated genre tokens across all catalog titles.
- International Movies, Dramas, and Comedies represent the top popular categories.

## Q3: Age Rating

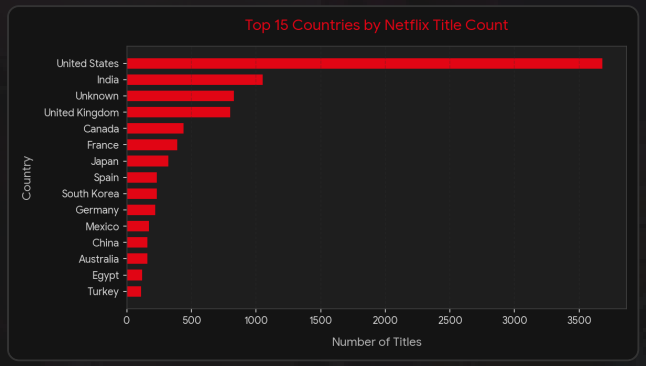
**Which age-rating category has the most content?**

- Evaluated distribution across age classification tags.
- TV-MA and TV-14 categories dominate the overall audience distribution.

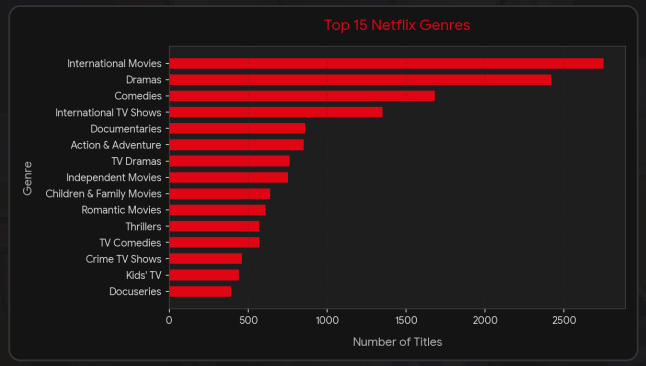


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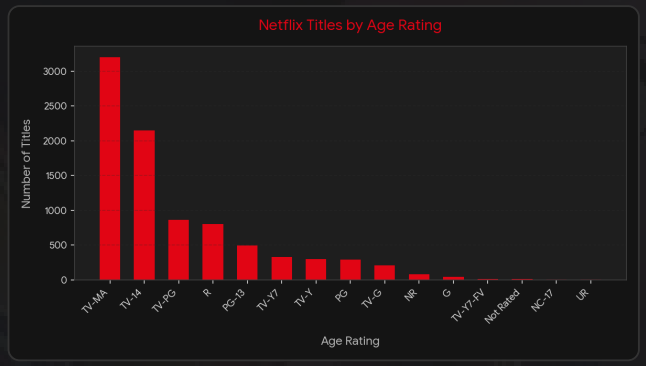
## 1. Country Production




## 2. Popular Genres




## 3. Age Rating Distribution






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# SUPERVISED MACHINE LEARNING MODELS



## 1. Linear SVC

- **Type:** Support Vector Classifier.
- **Function:** Finds optimal hyper-planes to separate multi-class targets.
- **Strengths:** Outperformed all models on rating classification, handling sparse high-dimensional features.

## 2. Logistic Regression

- **Type:** Linear Classification Model.
- **Function:** Predicts probabilities using decision boundaries.
- **Strengths:** Achieved highest accuracy and Macro F1 score on genre classification.

## 3. Random Forest

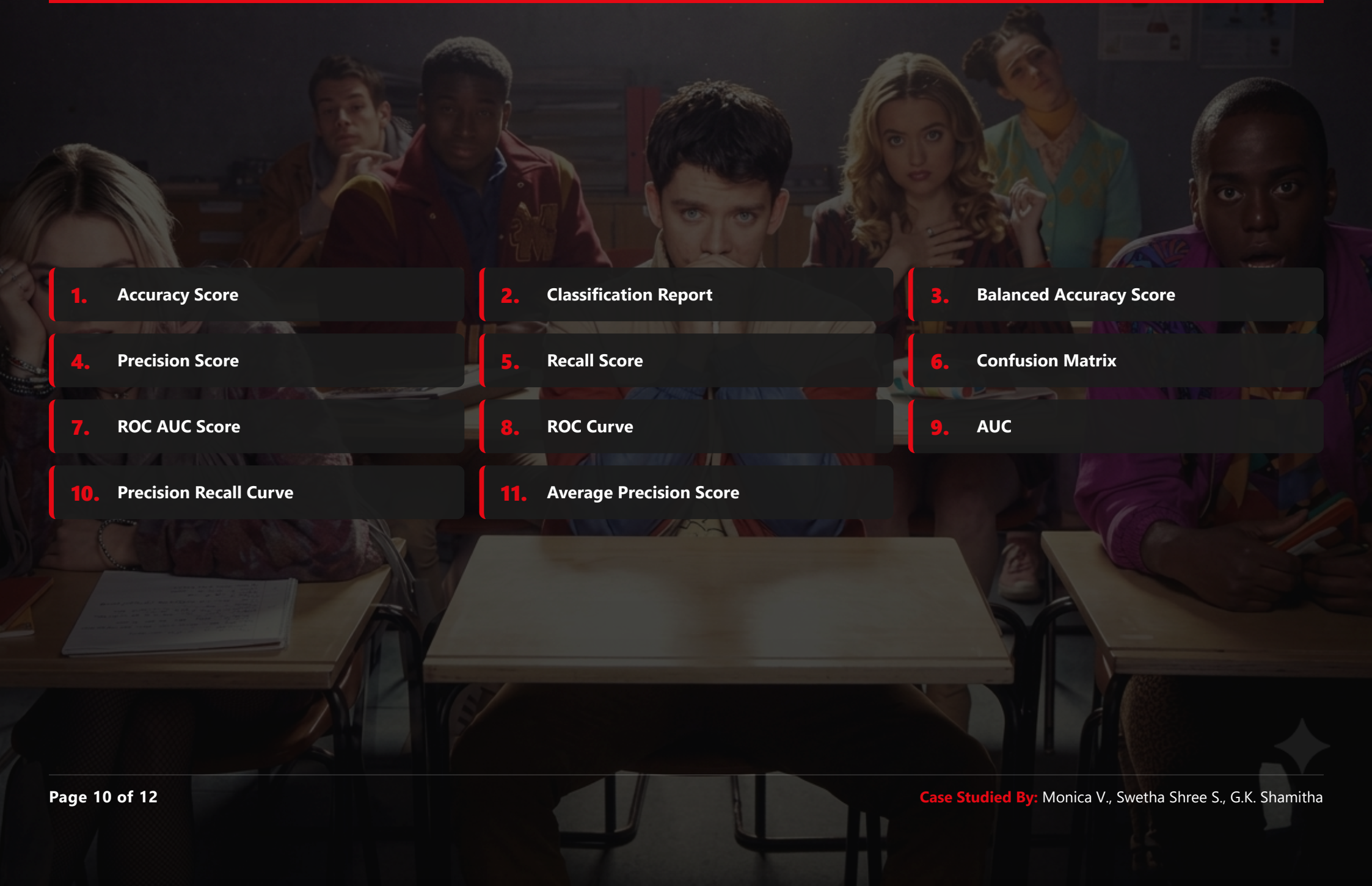
- **Type:** Ensemble Decision Tree Classifier.
- **Function:** Aggregates multiple randomized decision trees.
- **Strengths:** Delivered 100% accuracy on country classification task.



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# COMPREHENSIVE MODEL EVALUATION





1. Accuracy Score
2. Classification Report
3. Balanced Accuracy Score
4. Precision Score
5. Recall Score
6. Confusion Matrix
7. ROC AUC Score
8. ROC Curve
9. AUC
10. Precision Recall Curve
11. Average Precision Score



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# CONCLUSION & MODEL COMPARISON

Final Accuracy & Macro F1 Performance Results

## Model Accuracy Comparison Table

| Prediction Task | Logistic Regression | Random Forest | Linear SVM |
|---|---|---|---|
| **Country** | 99.94% | **100.00%** | **100.00%** |
| **Rating** | 49.88% | 42.12% | **50.71%** |
| **Genre** | **62.79%** | 51.31% | 61.76% |

## Best Model & Macro F1 Summary

| Prediction Target | Best Model | Macro F1 |
|---|---|---|
| **Country** | **Linear SVM / Random Forest** | 1.0000 |
| **Rating** | **Linear SVM** | 0.4774 |
| **Genre** | **Logistic Regression** | 0.6463 |

**Key Finding:** All three algorithms demonstrated strong classification capabilities. Comparatively, **Linear SVM** provided the highest overall predictive edge, achieving top accuracy on Rating and matching top performance on Country prediction.



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# USES AND APPLICATION

Practical Real-World Implementations & Deployment Opportunities

## 1. The Recommendation System (Cosine Similarity)

- **Real-life use:** This is exactly how Netflix, YouTube, or Amazon Prime suggest what you should watch next. If you finish a thriller show, the app immediately recommends other suspenseful thrillers to keep you on their app longer.

## 2. Categorizing New Content (Genre and Rating Classifiers)

- **Real-life use:** Instead of hiring humans to manually watch and tag thousands of new movies every day, streaming companies use these AI models to automatically organize and label catalogs instantly.

## 3. Regional Content Customization (Country Target Classifier)

- **Real-life use:** Streaming services can use this to customize their homepages based on where you live. For example, users in the US might see more US-based content, while international users get a custom homepage tailored to local productions.



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